

Aurora v1.2 RTL Architecture Specification

AUR-RTL-001 | Canonical Draft 1

1. Module Hierarchy

Module	Responsibility
aurora_core_top	Top-level core integration
aurora_front_end	Fetch, prefetch, branch prediction, extension assembly
aurora_issue_window	Four-entry decode window, selection, compaction
aurora_regfile	16 x 32 register file and writeback arbitration
aurora_lane0 / aurora_lane1	Symmetric general execution lanes
aurora_lsu	Load/store unit and address generation
aurora_mul_reduce	Multiplier, compressor and reduction resources
aurora_divider	Multi-cycle divider
aurora_stream_q0/q1	Stream state, staging, circular addressing
aurora_scoreboard	Busy state and completion tags
aurora_commit	In-order retirement and precise fault handling
aurora_interrupt	Priority arbitration and vector entry
aurora_memory_fabric	TCM/SRAM/MMIO/DMA/external arbitration

2. Interface Rules

- Ready/valid request-response interfaces between decoupled units
- Explicit age tags for issued/completion records
- No combinational path from external ready to fetch critical control
- Symmetric lane dispatch; shared resources selected by resource masks
- Pair destinations write both halves atomically

3. Clock and Reset

- Single core clock baseline; optional peripheral/timer clock domains
- Synchronous active-high internal reset preferred
- All cross-domain status signals synchronized
- Timer and DMA domains must not create variable interrupt semantics

4. Coding and Synthesis Rules

- SystemVerilog synthesizable subset
- No inferred latches
- Parameterized SRAM/TCM interfaces
- Assertions for protocol, ordering and pair-alias rules
- Generated constants from the machine-readable ISA database